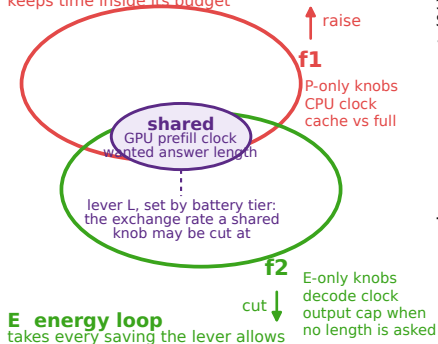


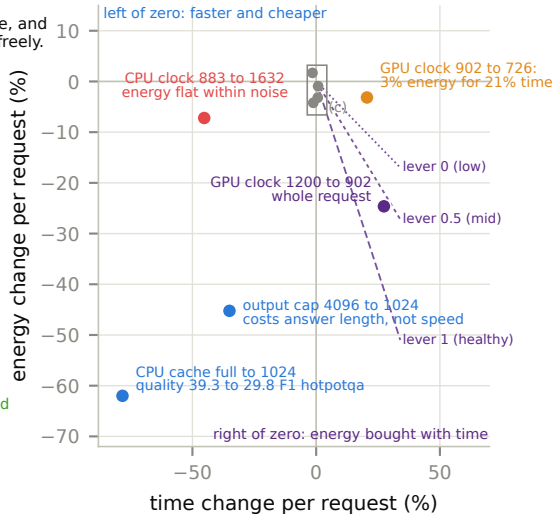
(a) two loops, one shared region

cutting f2 inside the shared region costs P one for one, and raising f1 there costs E. Outside it, each loop moves freely.

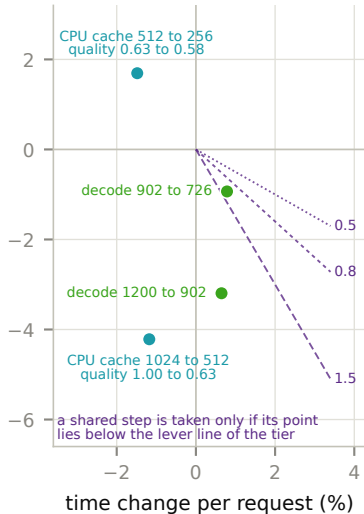
P performance loop
keeps time inside its budget



(b) the large steps



(c) the small steps



OnePlus 15, Llama-3.2-1B. GPU and split-clock steps from the cost table (9737 prompt + 1024 output tokens, Adreno 840); CPU clock from the 883 to 1632 MHz campaign; full cache from the six-generation CPU soak; output cap from the 4096 against 1024 token cells at 1200 MHz. Lever lines: energy saved per unit of time given up, by tier.